

Autonomous Driving Control & Planning Research Portfolio

Optimal Control | State Estimation | System Identification | Motion Planning | Real-Robot Validation

PRIMARY

Vehicle Control

Optimal control, trajectory tracking, system identification, state estimation, motion planning, and real-system validation.

SECONDARY

Planning & Prediction

Frenet/B-spline trajectory generation, obstacle and collision avoidance, vehicle-intent prediction, interaction reasoning, and simulation validation.

ROBOTICS & CONTROL FOUNDATION

B.S./M.S. concentration in Robotics & Control - UC Santa Cruz

Selected evidence

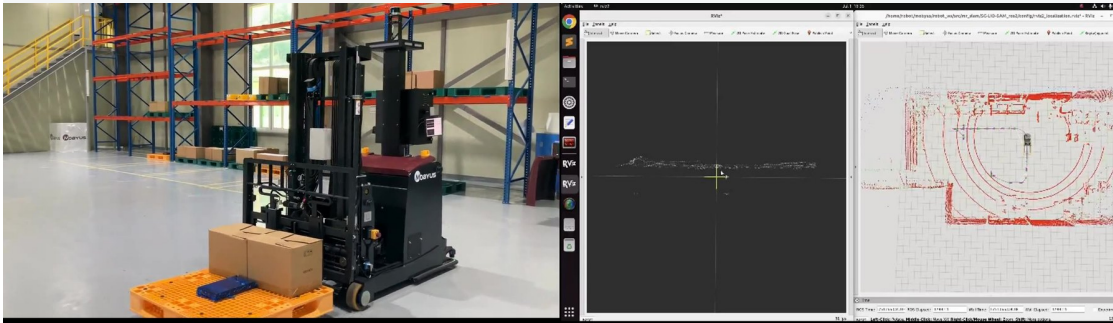
- Stochastic Dynamic Programming / HJB optimal-control policy over 34,596 states with 23/23 terminal-success validation tests.
- System identification using batch least squares and recursive least squares, plus EKF / nonlinear state estimation under uncertainty.
- Autonomous-forklift Frenet/B-spline motion planning, navigation integration, and real-robot path-following validation.
- Real-road trajectory prediction and critical-scenario reasoning using Continental radar/camera intersection data.

SCAN PORTFOLIO



MOBYUS FUTURE TECH R&D

Trajectory generation, path following, state estimation, docking, and closed-loop warehouse validation



Research value

This platform closes the loop from sensing and localization through trajectory generation, pallet perception, and physical validation, providing direct evidence of algorithm behavior outside offline datasets.

Control & planning scope

- Generated Frenet/B-spline trajectories and validated path following on an autonomous forklift in real warehouse conditions.
- Integrated ESKF-based LiDAR-IMU localization and ROS 2 navigation so trajectory generation receives stable state estimates.
- Connected obstacle avoidance, pallet docking, perception support, and PLC-facing workflows into the autonomy stack.
- Used calibration, log analysis, trajectory comparison, and failure diagnostics to reproduce and validate system behavior.

TRAJECTORY

Frenet / B-spline

CONTROLLED TESTBED

≤ 30 mm localization

Why it matters for Control

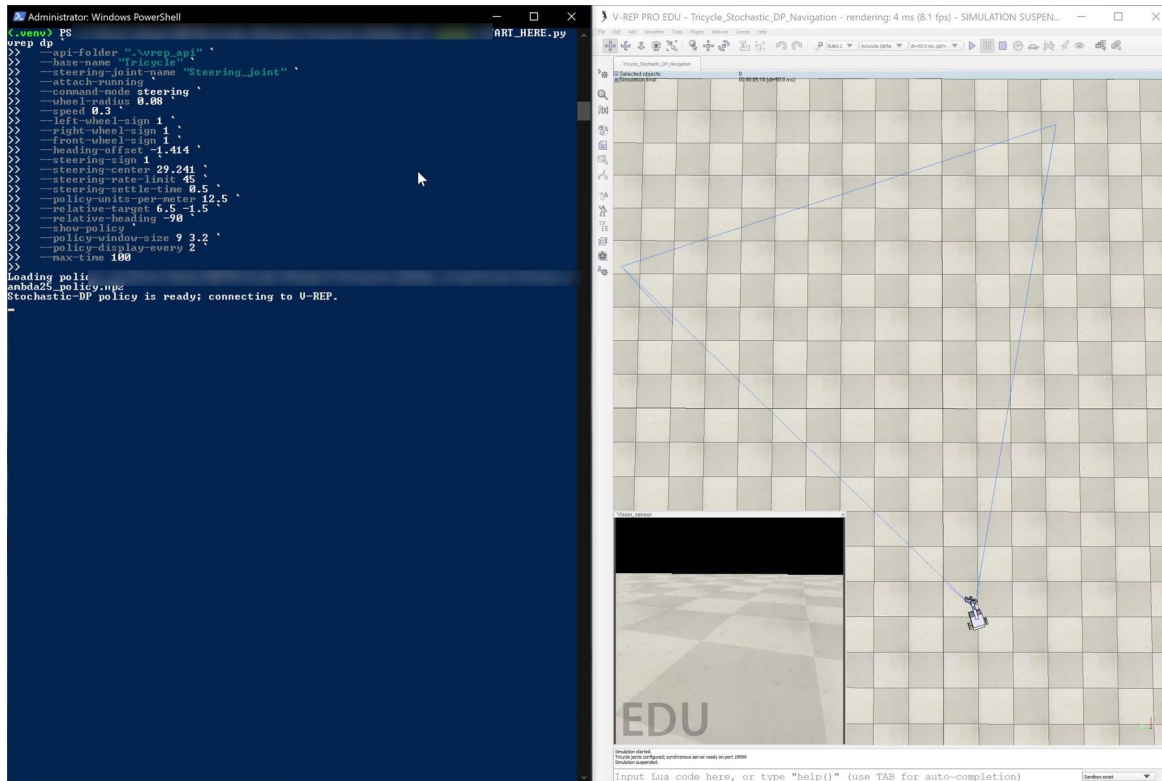
Demonstrates closed-loop integration from state estimation and trajectory generation through robot execution, debugging, and physical validation - without claiming passenger-car longitudinal/lateral control experience.

Robotics & Control Foundation

Optimal Control, State Estimation & System Identification

UC SANTA CRUZ

B.S./M.S. Robotics & Control | Autonomously Driving Tricycle with Monocular Vision - M.S. Project



Research pipeline

- Formulated stochastic Dynamic Programming / HJB value iteration using robot-frame relative position and terminal heading, with 11 discrete turn-rate actions and minimum-time / control-effort objectives.
- Computed a 34,596-state optimal policy and validated terminal success in 23/23 simulation tests.
- Designed EKF and second-order nonlinear estimators for pose, velocity, and heading from noisy monocular-camera position measurements; analyzed process / measurement covariance and model uncertainty.
- Performed system identification in V-REP/MATLAB using batch least squares and recursive least squares for discrete/continuous dynamic models.
- Implemented single- and two-pose monocular calibration using projective geometry and least-squares estimation.

POLICY SIZE

34,596 states DP / HJB

VALIDATION

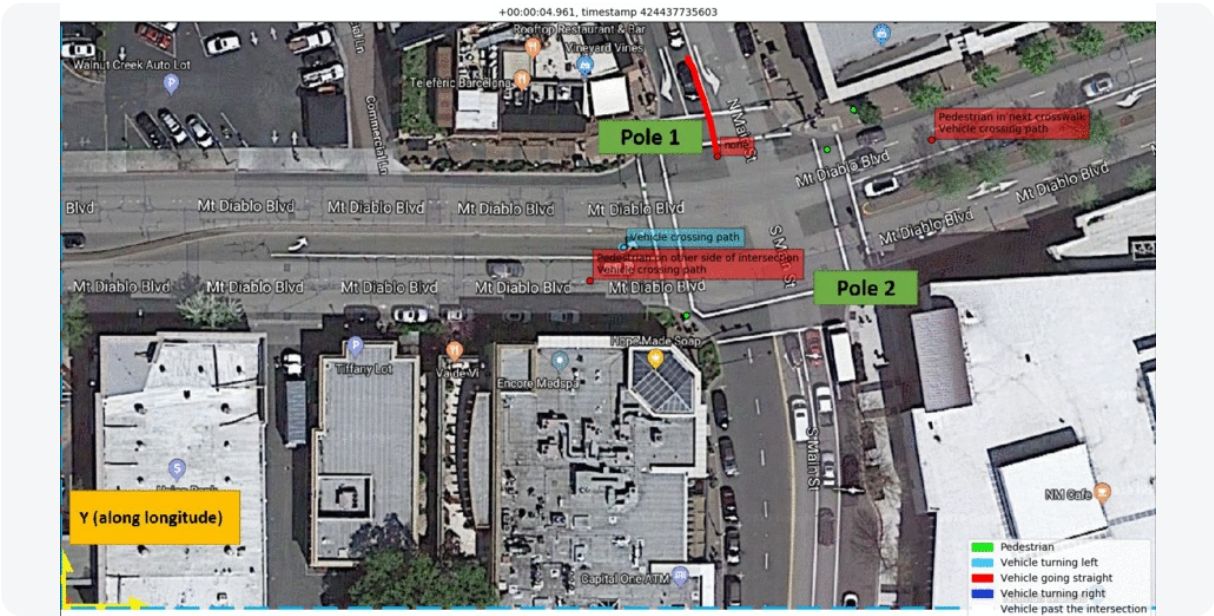
23 / 23 terminal success

Why it matters for Vehicle Control

Provides a mathematical foundation in optimal control, uncertainty-aware state estimation, and model identification that transfers directly to motion-control and trajectory-tracking problems.

CONTINENTAL-SPONSORED UCSC PROJECT

Radar/camera environment-model data from a real Walnut Creek intersection



Project scope

- Team research project using Continental-provided temporal radar data and camera footage from a real intelligent-intersection test site.
- Processed object trajectories using timestamp, object ID, x/y position, velocity, heading, object type, and time-in-intersection features.
- Built trajectory labeling and cleaning logic around crosswalk line-segment intersections to classify driver intent as left, straight, or right.
- Evaluated classical ML and LSTM sequence models for driver-intent / trajectory prediction and connected predictions to critical-scenario reasoning.
- Critical scenarios included vehicle-path crossings, vehicles entering the same exit direction/lane, and vehicle-pedestrian crosswalk conflicts.

REAL DATA PIPELINE

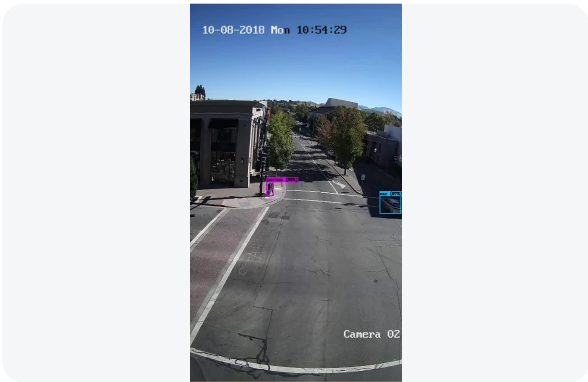
~15M

clean datapoints reported

TEAM LSTM RESULT

89.5 / 98.6%

before / after intersection



Why it matters for Planning & Prediction

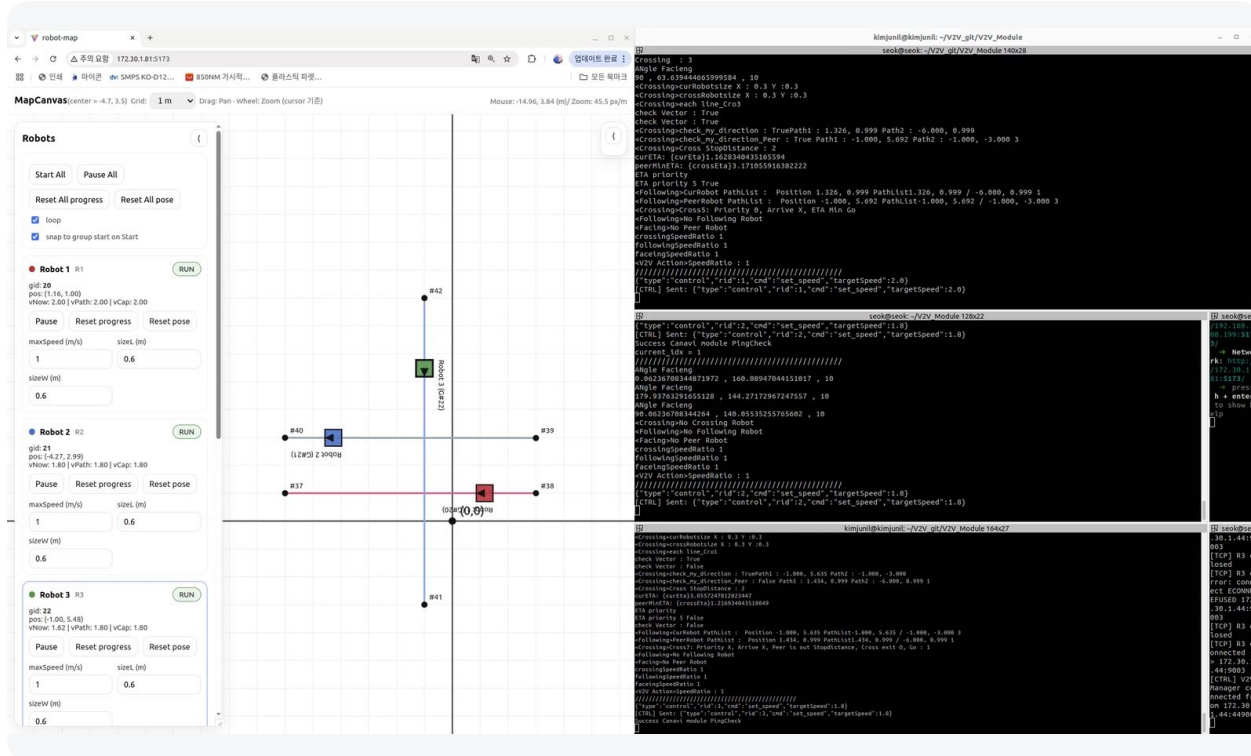
Direct experience with real intersection sensor data, temporal trajectory preprocessing, driver-intent / trajectory prediction, and safety-oriented conflict reasoning.

Interaction-Aware Motion Coordination & Collision Avoidance

PLANNING & CONTROL

V2V R&D

Trajectory-conflict reasoning, ETA / priority coordination, and simulation validation in shared regions



V2V multi-robot collision avoidance

- Exchanged peer robot pose/path information for crossing, following, and facing interactions.
- Reasoned about path intersections and shared occupancy/conflict regions.
- Used ETA / priority and safety margins to coordinate per-robot target speed around conflict areas.
- Validated behavior in simulation; public portfolio shows the demo while implementation source remains private.

Related patent

KR application 10-2024-0041376 - V2V-based collision avoidance for autonomous mobile robots (pending / unexamined).

Co-inventor. The simulation is public; implementation source remains private.